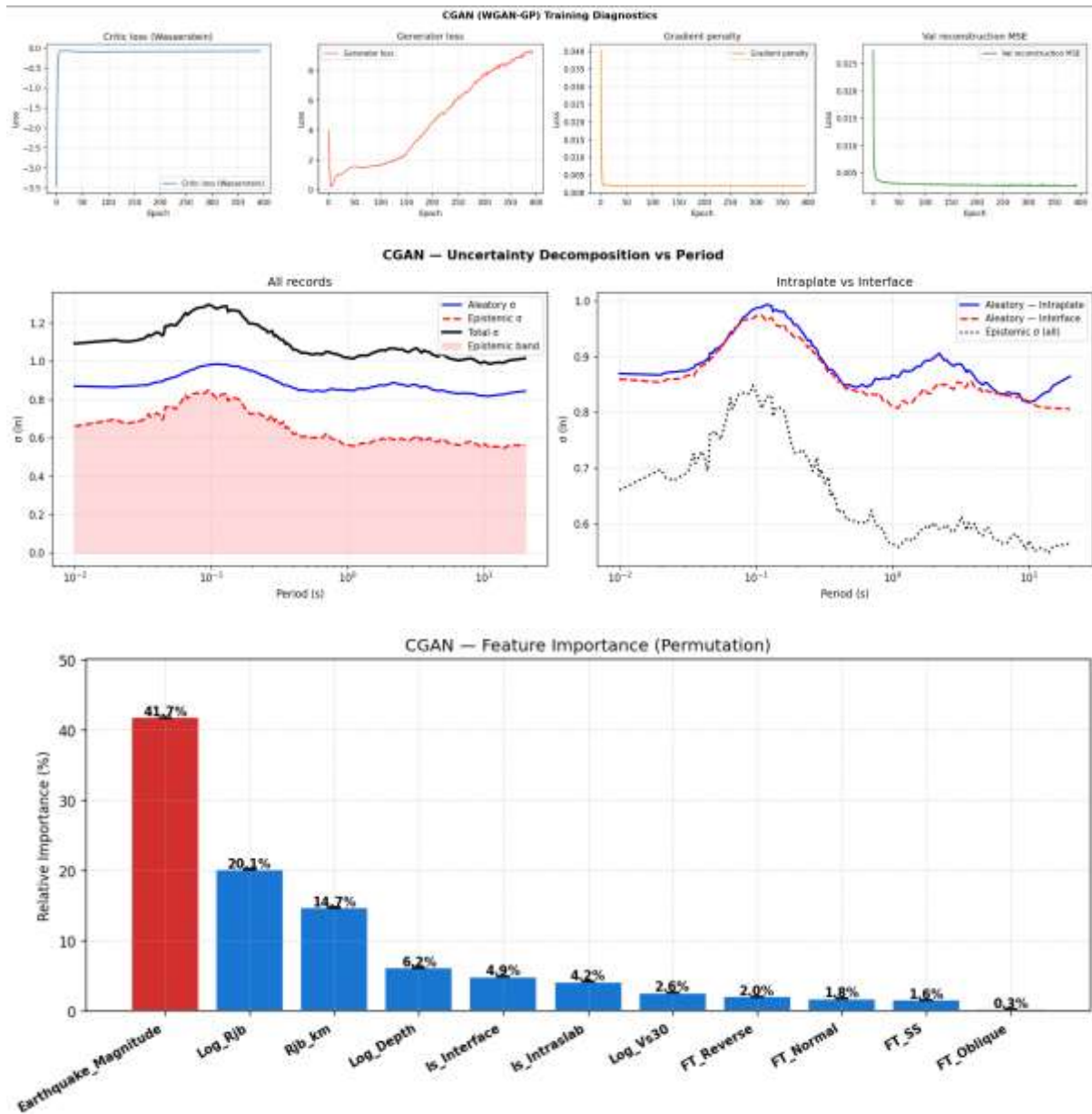
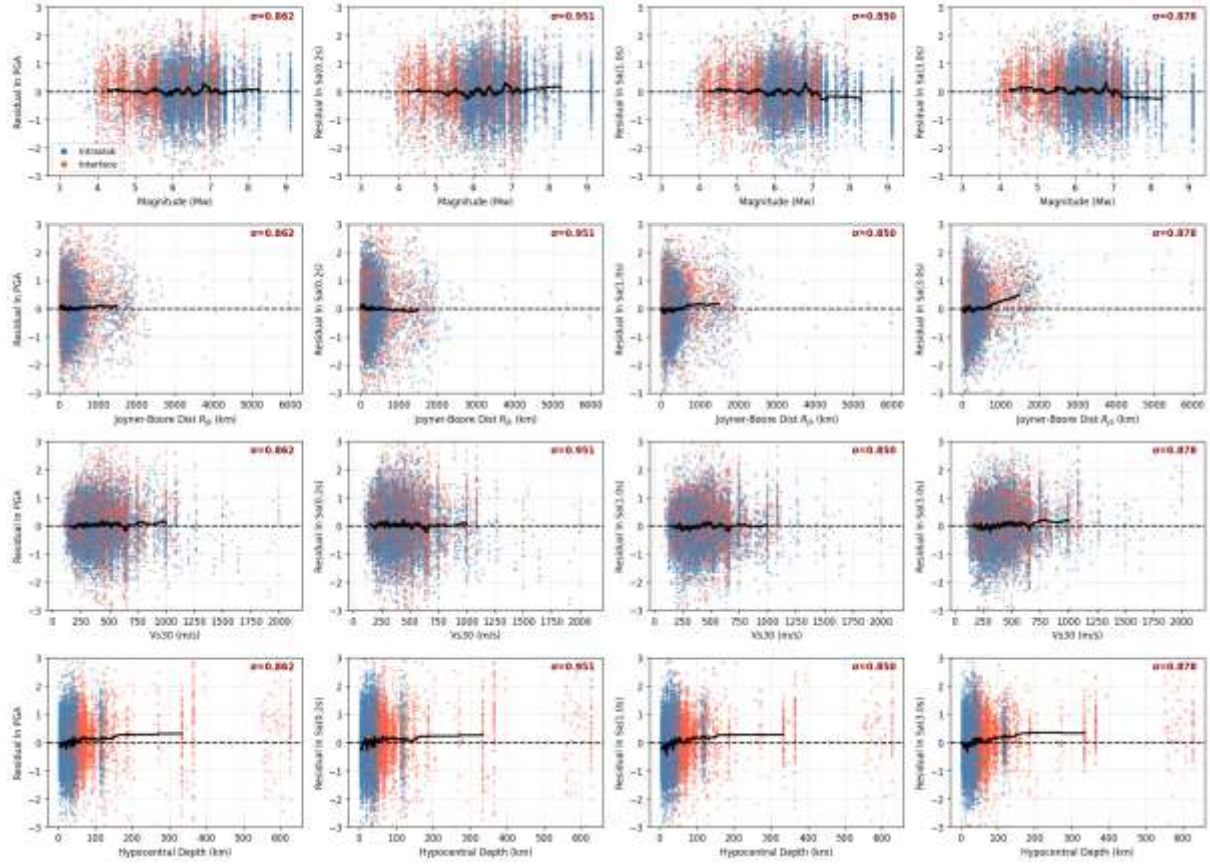


WGAN-GP Model Report, CE23B005, VISHNU VARDHAN

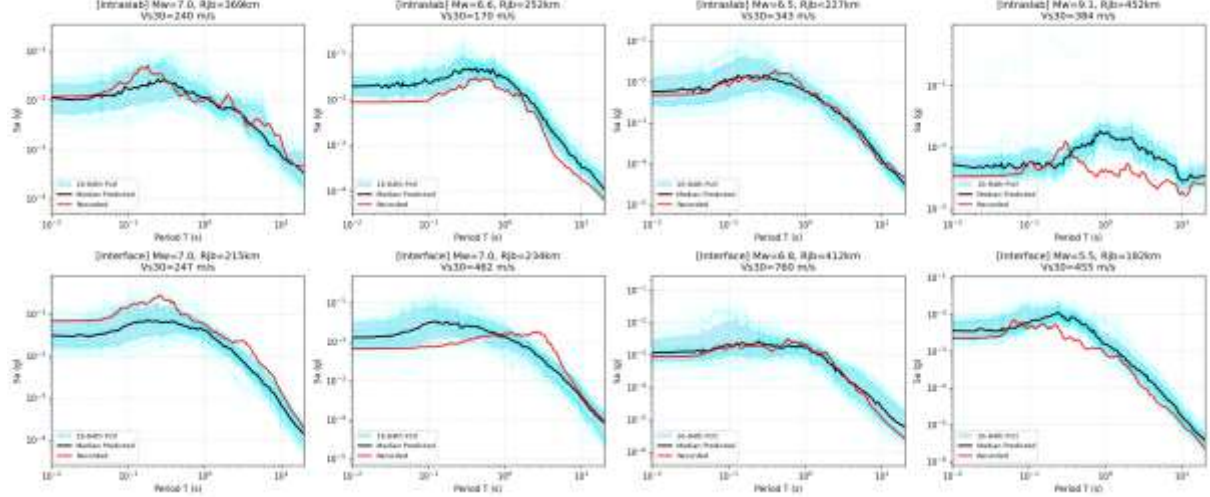


WGAN-GP — Residual Analysis (In scale)

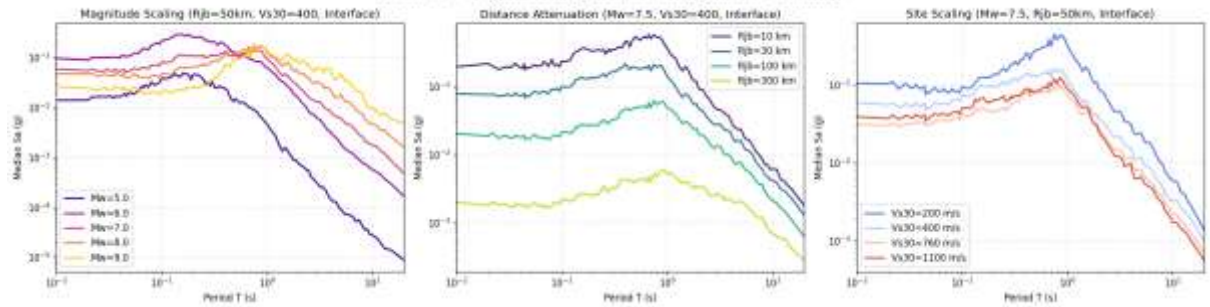


WGAN-GP — 100 Stochastic Realizations vs Recorded Spectra

Top: Intralab | Bottom: Interface



WGAN-GP — Physical Scaling Verification (Synthetic Scenarios)



1. Introduction

This project explores the use of a **Wasserstein Generative Adversarial Network with Gradient Penalty (WGAN-GP)** for predicting multiple ground motion parameters in a high-dimensional setting. The primary objective is to accurately model complex, non-linear relationships between input features and a large number of output variables, while also incorporating uncertainty estimation into the predictions.

Traditional regression models often struggle with such high-dimensional outputs and complex distributions. To overcome this, WGAN-GP is employed due to its improved stability compared to standard GANs. The use of the Wasserstein loss function ensures smoother gradients, while the gradient penalty enforces Lipschitz continuity, preventing issues like mode collapse and unstable training. This makes the model particularly suitable for engineering and scientific applications where both accuracy and reliability are critical.

2. Dataset Description

The dataset used in this project initially contained approximately 71,340 records. After thorough preprocessing and data cleaning, including the removal of missing, inconsistent, and noisy entries, a final dataset of 65,562 samples was obtained. This cleaned dataset provided a strong foundation for model training and evaluation.

Each data sample consists of 11 input features and 113 output variables. The input features likely represent parameters influencing ground motion (such as seismic or environmental factors), while the outputs correspond to various ground motion measures, including peak ground acceleration (PGA) and spectral accelerations at different time periods.

The high dimensionality of the output space makes this problem particularly challenging, as the model must learn relationships across a large number of dependent variables simultaneously. Proper preprocessing ensured that the data was normalized and structured appropriately, enabling efficient training and improved model performance.

3. Model and Methodology

The core model used in this project is the WGAN-GP architecture, implemented using TensorFlow. The model consists of two neural networks:

- A **generator**, which learns to produce realistic output samples given input features
- A **discriminator (critic)**, which evaluates the quality of generated samples

Unlike traditional GANs, WGAN-GP uses the Wasserstein distance as a loss function, which provides more meaningful gradients during training. Additionally, the gradient penalty term enforces constraints on the critic, leading to more stable and reliable training.

The model was trained for a maximum of 500 epochs. However, early stopping was applied at epoch 394 when the validation performance stopped improving. This helped prevent overfitting and ensured better generalization to unseen data. The best validation mean squared error (MSE) achieved during training was **0.002494**, indicating that the model successfully learned the underlying data distribution.

4. Observations and Results

The training process exhibited stable and consistent behaviour. The generator and discriminator losses followed expected patterns, with neither network overpowering the other, which is a key indicator of successful GAN training. The inclusion of gradient penalty significantly contributed to this stability.

The validation MSE showed a steady decline over epochs, demonstrating continuous learning and improvement. The use of early stopping ensured that the model retained optimal performance without overfitting. From a modelling perspective, the WGAN-GP architecture proved effective in handling the complexity of the dataset.

Residual analysis revealed that the prediction errors were centered around zero, indicating that the model does not suffer from systematic bias. However, a slight increase in error spread was observed for larger values of the outputs. This is a common phenomenon in regression tasks, especially when dealing with wide-ranging or highly variable data.

In terms of performance metrics, the model delivered strong results on the test dataset. The **R² score of 0.9069** indicates that approximately 91% of the variance in the output variables is explained by the model, which reflects a high level of accuracy. The **root mean squared error (RMSE) of 0.8843** and **mean absolute error (MAE) of 0.6398** are relatively low, confirming that the predicted values are close to the actual values. The **mean absolute percentage error (MAPE) of 25.95%** is moderately high but can be attributed to differences in output scales and the presence of smaller values in the dataset.

A detailed analysis of specific outputs such as PGA and spectral accelerations at different time periods showed strong correlation values exceeding 0.91. This demonstrates that the model performs consistently across different output variables and does not disproportionately favor any particular target.

An important strength of this model is its ability to quantify uncertainty. The model estimates both **aleatory uncertainty**, which represents inherent randomness in the data, and **epistemic uncertainty**, which reflects uncertainty in the model itself. The total uncertainty values ranged approximately from 0.986 to 1.296. This capability is particularly valuable in real-world applications such as earthquake engineering, where understanding the confidence of predictions is as important as the predictions themselves.

5. Conclusion

Overall, the WGAN-GP model demonstrates strong performance in handling a complex, high-dimensional multi-output prediction task. The model achieves high accuracy, as indicated by the R² score, and maintains low error metrics, confirming its reliability. The stable training process further validates the effectiveness of the WGAN-GP architecture.

One of the key highlights of this project is the integration of uncertainty estimation, which enhances the practical applicability of the model in risk-sensitive domains. The ability to provide both predictions and confidence measures makes this approach highly valuable for engineering and scientific decision-making.